

5. Write a Python function to analyze the histogram of the given input image based on color levels using open cv.

Code:

```
import cv2

import matplotlib.pyplot as plt

def analyze_histogram(image_path):

    # Read the image

    image = cv2.imread(image_path)

    if image is None:

        print("Error: Image not found!")

        return

    # Split into BGR channels

    channels = ('b', 'g', 'r')

    # Calculate and plot histogram for each channel

    for i, color in enumerate(channels):

        hist = cv2.calcHist([image], [i], None, [256], [0, 256])

        plt.plot(hist, color=color)

        plt.xlim([0, 256])

    # Labels and title

    plt.title("Color Histogram Analysis")

    plt.xlabel("Pixel Intensity (0-255)")

    plt.ylabel("Number of Pixels")

    plt.show()

# Function call

analyze_histogram(r'C:\Users\pbr22\OneDrive\Desktop\cv_lab\iphone.jpg')
```

Original Image:



Output:



